

AI ASSISTED CODING

LAB-15.5

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Task 1 – Student Attendance API

Task:

Create a RESTful API for student attendance tracking.

Instructions:

- **Endpoints required:**
 - o **GET /attendance → View attendance records**
 - o **POST /attendance → Mark attendance**
 - o **PUT /attendance/{id} → Update attendance record**
 - o **DELETE /attendance/{id} → Remove attendance entry**
- **Ensure JSON-based request and response handling.**

Expected Output:

- **API for maintaining and updating attendance records.**

PROMPT:

"""Generate a Python Flask REST API for student attendance with in-memory storage.

Implement GET /attendance, POST /attendance, PUT /attendance/<id>, DELETE /attendance/<id>.

Use JSON request/response and add comments."""

GIVEN CODE:

```

lab15.py > create_attendance
1  """Generate a Python Flask REST API for student attendance with in-memory storage.
2  Implement GET /attendance, POST /attendance, PUT /attendance/<id>, DELETE /attendance/<id>.
3  Use JSON request/response and add comments."""
4  from flask import Flask, request, jsonify
5  app = Flask(__name__)
6  # In-memory storage for attendance records
7  attendance_records = []
8  # GET /attendance - Retrieve all attendance records
9  @app.route('/attendance', methods=['GET'])
10 def get_attendance():
11     return jsonify(attendance_records), 200
12 # POST /attendance - Create a new attendance record
13 @app.route('/attendance', methods=['POST'])
14 def create_attendance():
15     data = request.get_json()
16     if not data or 'student_name' not in data or 'date' not in data:
17         return jsonify({'error': 'Invalid input'}), 400
18     new_record = {}
19     'id': len(attendance_records) + 1,
20     'student_name': data['student_name'],
21     'date': data['date']
22 }
23 attendance_records.append(new_record)
24 return jsonify(new_record), 201
25 # PUT /attendance/<id> - Update an existing attendance record
26 @app.route('/attendance/<int:record_id>', methods=['PUT'])
27 def update_attendance(record_id):
28     data = request.get_json()
29     if not data or 'student_name' not in data or 'date' not in data:
30         return jsonify({'error': 'Invalid input'}), 400
31     for record in attendance_records:
32         if record['id'] == record_id:
33             record['student_name'] = data['student_name']
34             record['date'] = data['date']
35             return jsonify(record), 200
36     return jsonify({'error': 'Record not found'}), 404
37 # DELETE /attendance/<id> - Delete an attendance record
38 @app.route('/attendance/<int:record_id>', methods=['DELETE'])
39 def delete_attendance(record_id):
40     global attendance_records
41     attendance_records = [record for record in attendance_records if record['id'] != record_id]
42     return jsonify({'message': 'Record deleted'}), 200
43 if __name__ == '__main__':
44     app.run(debug=True)
45

```

Task 2 – Inventory Management API

Task:

Develop a REST API for warehouse inventory management.

Instructions:

- Endpoints required:
 - o GET /inventory → List all items
 - o POST /inventory → Add new item
 - o PUT /inventory/{id} → Update stock quantity
 - o DELETE /inventory/{id} → Remove an item
- Implement validation for stock values.

Expected Output:

- API capable of managing inventory stock levels.

PROMPT:

"""Create a Flask REST API for inventory management with item id, name, and quantity.

Implement GET, POST, PUT, DELETE endpoints for /inventory.

Add validation so stock quantity cannot be negative."""

GIVEN CODE AND OUTPUT:

```
lab15.py
1  """Create a Flask REST API for inventory management with item id, name, and quantity.
2  Implement GET, POST, PUT, DELETE endpoints for /inventory.
3  Add validation so stock quantity cannot be negative."""
4  from flask import Flask, request, jsonify
5  app = Flask(__name__)
6  # In-memory storage for inventory items
7  inventory = {}
8  @app.route('/inventory', methods=['GET'])
9  def get_inventory():
10     """Endpoint to retrieve the entire inventory."""
11     return jsonify(inventory)
12  @app.route('/inventory', methods=['POST'])
13  def add_item():
14     """Endpoint to add a new item to the inventory."""
15     data = request.get_json()
16     item_id = data.get('id')
17     name = data.get('name')
18     quantity = data.get('quantity')
19
20     # Validate input
21     if not isinstance(item_id, int) or not isinstance(name, str) or not isinstance(quantity, int):
22         return jsonify({"error": "Invalid input: id must be an integer, name must be a string, and quantity must be an integer."}), 400
23     if quantity < 0:
24         return jsonify({"error": "Invalid input: quantity cannot be negative."}), 400
25
26     # Add item to inventory
27     inventory[item_id] = {"name": name, "quantity": quantity}
28     return jsonify({"message": "Item added successfully."}), 201
29  @app.route('/inventory/<int:item_id>', methods=['PUT'])
30  def update_item(item_id):
31     """Endpoint to update an existing item in the inventory."""
32     if item_id not in inventory:
33         return jsonify({"error": "Item not found."}), 404
34
35     data = request.get_json()
36     name = data.get('name')
37     quantity = data.get('quantity')
38
39     # Validate input
40     if name is not None and not isinstance(name, str):
41         return jsonify({"error": "Invalid input: name must be a string."}), 400
42     if quantity is not None and not isinstance(quantity, int):
43         return jsonify({"error": "Invalid input: quantity must be an integer."}), 400
44     if quantity < 0:
45         return jsonify({"error": "Invalid input: quantity cannot be negative."}), 400
46
47     # Update item
48     inventory[item_id]['name'] = name
49     inventory[item_id]['quantity'] = quantity
50     return jsonify({"message": "Item updated successfully."}), 200
51
52  if __name__ == '__main__':
53     app.run(debug=True)
```

PROBLEMS OUTPUT DEBUG CONSOLE **TERMINAL** PORTS GIT LENS POSTMAN CONSOLE

```
* Debugger PIN: 615-148-475
PS C:\Users\katta\OneDrive\Desktop\AIAC> & C:/Users/katta/anaconda3/python.exe c:/Users/katta/OneDrive/Desktop/AIAC/lab15.py
* Serving Flask app 'lab15'
* Debug mode: on
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on http://127.0.0.1:5000
Press CTRL+C to quit
* Restarting with watchdog (windowsapi)
* Debugger is active!
* Debugger PIN: 615-148-475
```

Task 3 – Hotel Room Booking API

Task:

Develop a RESTful API for hotel room booking.

Instructions:

- Endpoints required:

- o GET /rooms → View available rooms
- o POST /rooms/book → Book a room
- o GET /rooms/{id} → View booking details
- o DELETE /rooms/{id} → Cancel booking
- Implement error handling for unavailable rooms.

Expected Output:

- API simulating hotel room booking operations.

PROMPT:

"""Generate a Flask REST API for hotel room booking using in-memory data.

Endpoints: GET /rooms, POST /rooms/book, GET /rooms/<id>, DELETE /rooms/<id>.

Add error handling if a room is already booked."""

GIVEN OUTPUT:

```

1  """Generate a Flask REST API for hotel room booking using in-memory data.
2  Endpoints: GET /rooms, POST /rooms/book, GET /rooms/<id>, DELETE /rooms/<id>.
3  Add error handling if a room is already booked."""
4  from flask import Flask, request, jsonify
5  app = Flask(__name__)
6  # In-memory storage for rooms
7  rooms = {
8      1: {"type": "single", "price": 100, "booked": False},
9      2: {"type": "double", "price": 150, "booked": False},
10     3: {"type": "suite", "price": 300, "booked": False}
11 }
12 @app.route('/rooms', methods=['GET'])
13 def get_rooms():
14     """Endpoint to get the list of all rooms with their details."""
15     return jsonify(rooms)
16 @app.route('/rooms/book', methods=['POST'])
17 def book_room():
18     """Endpoint to book a room. Expects JSON with 'id' of the room to be booked."""
19     data = request.get_json()
20     room_id = data.get('id')
21     if room_id not in rooms:
22         return jsonify({"error": "Room not found"}), 404
23     if rooms[room_id]['booked']:
24         return jsonify({"error": "Room already booked"}), 400
25     rooms[room_id]['booked'] = True
26     return jsonify({"message": f"Room {room_id} booked successfully"})
27 @app.route('/rooms/<int:room_id>', methods=['GET'])
28 def get_room(room_id):
29     """Endpoint to get details of a specific room by its ID."""
30     if room_id not in rooms:
31         return jsonify({"error": "Room not found"}), 404
32     return jsonify(rooms[room_id])
33 @app.route('/rooms/<int:room_id>', methods=['DELETE'])
34 def delete_room(room_id):
35     """Endpoint to delete a room by its ID. This will remove the room from the in-memory storage."""
36     if room_id not in rooms:
37         return jsonify({"error": "Room not found"}), 404
38     del rooms[room_id]
39     return jsonify({"message": f"Room {room_id} deleted successfully"})
40 if __name__ == '__main__': app.run(debug=True)

```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS POSTMAN CONSOLE

```

* Debug mode: on
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on http://127.0.0.1:5000
Press CTRL+C to quit
* Restarting with watchdog (windowsapi)
* Debugger is active!
* Debugger PIN: 615-148-475
* Detected change in 'c:\\Users\\katta\\OneDrive\\Desktop\\AIAC\\lab15.py', reloading
* Detected change in 'c:\\Users\\katta\\OneDrive\\Desktop\\AIAC\\lab15.py', reloading
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* Debugger is active!
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```

Task 4 – Online Voting API

Task:

Create a RESTful API for an online voting system.

Instructions:

- Endpoints required:
 - o GET /candidates → List candidates
 - o POST /vote → Cast a vote
 - o GET /results → View voting results
- Ensure one vote per user using in-memory validation.

Expected Output:

- API simulating a secure online voting process.

PROMPT:

"""Create a Flask REST API for an online voting system.

Endpoints: GET /candidates, POST /vote, GET /results.

Ensure each user can vote only once using in-memory validation."""

GIVEN OUTPUT:

```
lab15.py > ...
1  """Create a Flask REST API for an online voting system.
2  Endpoints: GET /candidates, POST /vote, GET /results.
3  Ensure each user can vote only once using in-memory validation."""
4  from flask import Flask, request, jsonify
5  app = Flask(__name__)
6  # In-memory storage for candidates and votes
7  candidates = ["Alice", "Bob", "Charlie"]
8  votes = {}
9  @app.route('/candidates', methods=['GET'])
10 def get_candidates():
11     """Endpoint to retrieve the list of candidates."""
12     return jsonify(candidates)
13 @app.route('/vote', methods=['POST'])
14 def cast_vote():
15     """Endpoint to cast a vote for a candidate."""
16     data = request.get_json()
17     user_id = data.get('user_id')
18     candidate = data.get('candidate')
19     # Validate input
20     if not user_id or not candidate:
21         return jsonify({"error": "User ID and candidate are required."}), 400
22     if candidate not in candidates:
23         return jsonify({"error": "Invalid candidate."}), 400
24     if user_id in votes:
25         return jsonify({"error": "User has already voted."}), 400
26     # Cast the vote
27     votes[user_id] = candidate
28     return jsonify({"message": "Vote cast successfully."}), 200
29 @app.route('/results', methods=['GET'])
30 def get_results():
31     """Endpoint to retrieve the voting results."""
32     results = {candidate: 0 for candidate in candidates}
33     for vote in votes.values():
34         if vote in results:
35             results[vote] += 1
36     return jsonify(results)
37 if __name__ == '__main__':
38     app.run(debug=True)
39 #TESTING
40 # To test the API, you can use tools like Postman or curl to send requests to the endpoints:
41 # 1. Get candidates: curl http://localhost:5000/candidates
42 # 2. Cast a vote: curl -X POST http://localhost:5000/vote -H "Content-Type: application/json" -d '{"user_id": "user1", "candidate": "Alice"}'
43 # 3. Get results: curl http://localhost:5000/results

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS POSTMAN CONSOLE
Press CTRL+C to quit
* Restarting with watchdog (windowsapi)
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* Debugger PIN: 615-148-475
```

Task 5 – Teacher Management API

Task:

Develop a RESTful API to manage teacher records.

Instructions:

- Endpoints required:
- o GET /teachers → List all teachers

- o POST /teachers → Add a new teacher
- o PUT /teachers/{id} → Update teacher details
- o DELETE /teachers/{id} → Remove a teacher
- Ensure proper error handling and JSON responses.

Expected Output:

- Functional API for managing teacher information.

PROMPT:

"""Generate a Flask REST API to manage teachers with id, name, and subject.

Implement GET, POST, PUT, DELETE endpoints for /teachers.

Return JSON responses and handle errors properly."""

GIVEN OUTPUT:

```

"""Generate a Flask REST API to manage teachers with id, name, and subject.
Implement GET, POST, PUT, DELETE endpoints for /teachers.
Return JSON responses and handle errors properly."""
from flask import Flask, request, jsonify

app = Flask(__name__)
# In-memory storage for teachers
teachers = {}
@app.route('/teachers', methods=['GET'])
def get_teachers():
    """Endpoint to retrieve all teachers."""
    return jsonify(list(teachers.values())), 200
@app.route('/teachers', methods=['POST'])
def add_teacher():
    """Endpoint to add a new teacher."""
    data = request.get_json()
    if not data or 'name' not in data or 'subject' not in data:
        return jsonify({'error': 'Invalid input'}), 400
    teacher_id = len(teachers) + 1
    teacher = {
        'id': teacher_id,
        'name': data['name'],
        'subject': data['subject']
    }
    teachers[teacher_id] = teacher
    return jsonify(teacher), 201
@app.route('/teachers/<int:teacher_id>', methods=['PUT'])
def update_teacher(teacher_id):
    """Endpoint to update an existing teacher."""
    if teacher_id not in teachers:
        return jsonify({'error': 'Teacher not found'}), 404
    data = request.get_json()
    if not data or 'name' not in data or 'subject' not in data:
        return jsonify({'error': 'Invalid input'}), 400
    teacher = teachers[teacher_id]
    teacher['name'] = data['name']
    teacher['subject'] = data['subject']
    return jsonify(teacher), 200
@app.route('/teachers/<int:teacher_id>', methods=['DELETE'])
def delete_teacher(teacher_id):
    """Endpoint to delete a teacher."""
    if teacher_id not in teachers:
        return jsonify({'error': 'Teacher not found'}), 404
    del teachers[teacher_id]
    return jsonify({'message': 'Teacher deleted'}), 200

```

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Press CTRL+C to quit

Restarting with watchdog (windowsapi)

Debugger is active!

Debugger PIN: 615-148-475

Detected change in 'c:\\Users\\katta\\OneDrive\\Desktop\\AIAC\\lab15.py', reloading

Detected change in 'c:\\Users\\katta\\OneDrive\\Desktop\\AIAC\\lab15.py', reloading

Restarting with watchdog (windowsapi)

Debugger is active!

Debugger PIN: 615-148-475